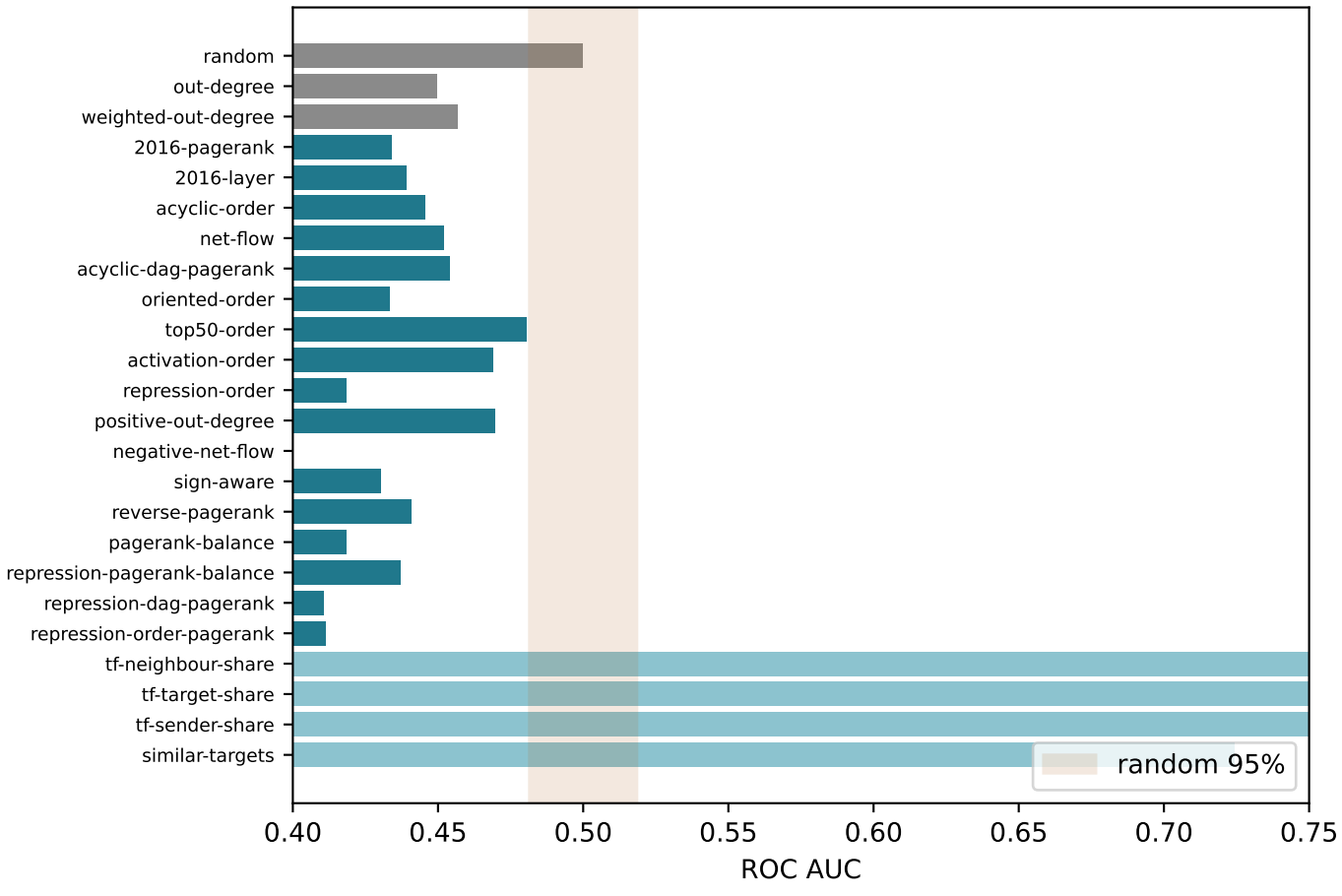
# Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,986 genes ranked, 1,156 in lambert-2018-tfs.txt · top k = 100

| Pipeline | Reference genes in top 100 | AUC | p top-k | p AUC | AUC vs best baseline (95%) | p vs baseline |
|---|---|---|---|---|---|---|
| random | 9.5 (4–16) | 0.500 (0.481–0.519) | | | | |
| out-degree | 5 | 0.450 | 0.967 | 1.000 | -0.007 (-0.008, -0.006) | < 0.001 |
| weighted-out-degree | 6 | 0.457 | 0.922 | 1.000 | +0.000 (+0.000, +0.000) | |
| 2016-pagerank | 9 | 0.434 | 0.620 | 1.000 | -0.023 (-0.034, -0.010) | < 0.001 |
| 2016-layer | 8 | 0.439 | 0.745 | 1.000 | -0.018 (-0.032, -0.003) | 0.016 |
| acyclic-order | 4 | 0.445 | 0.987 | 1.000 | -0.011 (-0.017, -0.005) | < 0.001 |
| net-flow | 4 | 0.452 | 0.987 | 1.000 | -0.005 (-0.014, +0.004) | 0.306 |
| acyclic-dag-pagerank | 6 | 0.454 | 0.922 | 1.000 | -0.003 (-0.008, +0.002) | 0.274 |
| oriented-order | 5 | 0.433 | 0.967 | 1.000 | -0.023 (-0.034, -0.013) | < 0.001 |
| top50-order | 6 | 0.481 | 0.922 | 0.980 | +0.024 (+0.005, +0.043) | 0.002 |
| activation-order | 11 | 0.469 | 0.352 | 1.000 | +0.012 (+0.005, +0.019) | < 0.001 |
| repression-order | 3 | 0.418 | 0.998 | 1.000 | -0.038 (-0.050, -0.027) | < 0.001 |
| positive-out-degree † | 9 | 0.470 | 0.620 | 1.000 | +0.013 (+0.010, +0.016) | < 0.001 |
| negative-net-flow † | 3 | 0.399 | 0.998 | 1.000 | -0.057 (-0.070, -0.045) | < 0.001 |
| sign-aware † | 4 | 0.430 | 0.987 | 1.000 | -0.026 (-0.033, -0.020) | < 0.001 |
| reverse-pagerank † | 3 | 0.441 | 0.998 | 1.000 | -0.016 (-0.019, -0.013) | < 0.001 |
| pagerank-balance † | 3 | 0.418 | 0.998 | 1.000 | -0.038 (-0.050, -0.026) | < 0.001 |
| repression-pagerank-balance † | 7 | 0.437 | 0.851 | 1.000 | -0.019 (-0.031, -0.008) | < 0.001 |
| repression-dag-pagerank † | 5 | 0.411 | 0.967 | 1.000 | -0.046 (-0.059, -0.033) | < 0.001 |
| repression-order-pagerank † | 4 | 0.411 | 0.987 | 1.000 | -0.045 (-0.056, -0.034) | < 0.001 |
| tf-neighbour-share ‡ | 99 | 0.769 | < 0.001 | < 0.001 | +0.312 (+0.289, +0.336) | < 0.001 |
| tf-target-share ‡ | 98 | 0.772 | < 0.001 | < 0.001 | +0.315 (+0.292, +0.340) | < 0.001 |
| tf-sender-share ‡ | 99 | 0.755 | < 0.001 | < 0.001 | +0.299 (+0.275, +0.322) | < 0.001 |
| similar-targets ‡ | 95 | 0.724 | < 0.001 | < 0.001 | +0.268 (+0.242, +0.292) | < 0.001 |

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



# Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

## collectri

| Graph | Pairs | Recall (×) | Positive × | Negative × | Sign (chance) |
|---|---|---|---|---|---|
| raw graph | 31,891 | 0.044 (2.59×) | 3.19× | 0.78× | 0.82 (0.81) |
| resolve_reciprocals → drop_sinks → prune_triangles | 30,803 | 0.004 (1.09×) | 3.65× | 0.80× | 0.40 (0.40) |
| resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering | 22,712 | 0.003 (0.82×) | 3.93× | 0.68× | 0.40 (0.34) |
| absolute | 31,891 | 0.044 (2.59×) | | | |
| absolute → minimum_feedback_order | 30,990 | 0.030 (2.31×) | | | |
| orient_reciprocals ratio=2.0 → absolute | 30,874 | 0.017 (1.78×) | | | |
| orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order | 30,874 | 0.016 (1.77×) | | | |
| top_outgoing k=50 → absolute | 31,891 | 0.016 (4.27×) | | | |
| top_outgoing k=50 → absolute → minimum_feedback_order | 30,876 | 0.011 (4.00×) | | | |
| keep_sign sign=1 → absolute | 31,891 | 0.041 (3.19×) | | | |
| keep_sign sign=1 → absolute → minimum_feedback_order | 30,789 | 0.027 (2.81×) | | | |
| keep_sign sign=-1 → absolute | 29,662 | 0.004 (0.79×) | | | |
| keep_sign sign=-1 → absolute → minimum_feedback_order | 29,143 | 0.003 (0.80×) | | | |
| keep_sign sign=-1 | 29,662 | 0.004 (0.79×) | | 0.79× | 0.21 (0.21) |

# Pipeline steps

### out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,634,936 edges (704,580 negative), largest strong component 11,983.

### weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,634,936 edges (704,580 negative), largest strong component 11,983.

### 2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 672,077 edges (614,240 negative), largest strong component 11,620.

### 2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 672,077 edges (614,240 negative), largest strong component 11,620; DAG keeps 388,790 edges.

### acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,634,936 edges (0 negative), largest strong component 11,983; DAG keeps 1,996,774 edges.

### net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,634,936 edges (0 negative), largest strong component 11,983.

### acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,634,936 edges (0 negative), largest strong component 11,983; DAG keeps 1,996,774 edges.

### oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,440,319 edges (0 negative), largest strong component 10,287; DAG keeps 1,428,905 edges.

### top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 550,595 edges (0 negative), largest strong component 11,982; DAG keeps 383,760 edges.

### activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,930,356 edges (0 negative), largest strong component 11,978; DAG keeps 1,393,607 edges.

### repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 704,580 edges (0 negative), largest strong component 11,017; DAG keeps 607,145 edges.

### positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,634,936 edges (704,580 negative), largest strong component 11,983.

### negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,634,936 edges (704,580 negative), largest strong component 11,983.

### sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,634,936 edges (704,580 negative), largest strong component 11,983.

### reverse-pagerank: Highest reverse PageRank (raw graph)